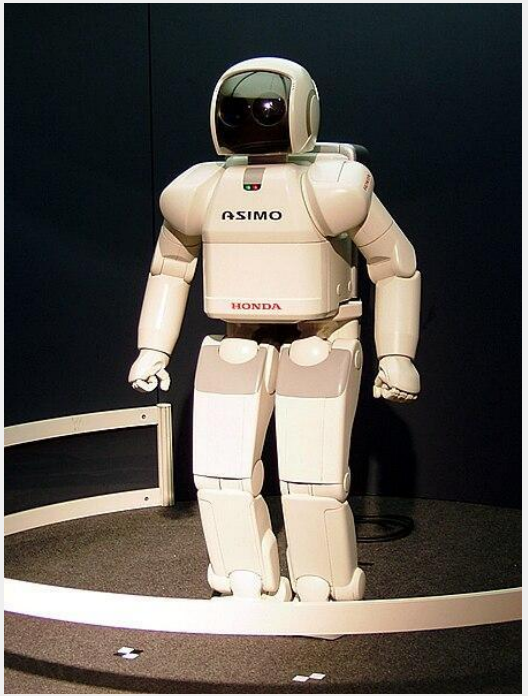
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 7 March 7, 2025

February 25, 2025 Global Affairs

Estonia and OpenAI to bring ChatGPT to schools nationwide

OpenAI to work with Estonia's government on the world's first initiative to integrate ChatGPT Edu into a national education system.



Called "AI Leap," the programme will initially include 20,000 high schoolers and 3,000 teachers from September 1, 2025, the education ministry said.

Estonia then hopes to expand to vocational schools and an additional 38,000 students and 3,000 teachers from September 2026.

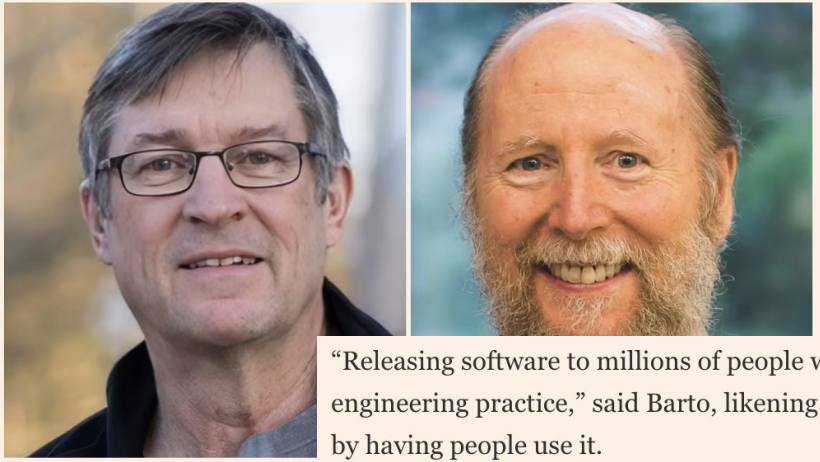
OpenAI is proud to work with Estonia's government on a world-first initiative to provide all students and teachers in the secondary school system with access to ChatGPT Edu, a

Artificial intelligence + Add to myFT

Turing Award winners warn over unsafe deployment of AI models

Two pioneers of reinforcement learning have won the $1mn prize from the Association for Computing Machinery

Save



Andrew Barto, left, and Richard Sutton are concern them

Cristina Criddle in San Francisco and **Meliss**

Published 7 HOURS AGO

"Releasing software to millions of people without safeguards is not good engineering practice," said Barto, likening it to building a bridge and testing it by having people use it.

"Engineering practice has evolved to try to mitigate the negative consequences of technology, and I don't see that being practised by the companies that are developing," he added.

Richard Sutton &
Andrew Barto

AI History



MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
					1	2
3	4	5	6	7 remote class + guest lecture	8	9
10	11	12	13	14 normal class ✓		16
17	18	19	20	21 normal class ✓		23
24	25	26	27	28 midterm	29	30
31				spring break		

Last Week!

- Some terms: Effective altruism, singularity, posthumanism, AI safety, AI alignment, superintelligence
- How did finance and ideology fuel Silicon Valley investment in AI?
- Is AI just a big tech game?

NOT SO FAST



STEPHEN HAWKING
Not afraid of black holes. A.I. is another story.



BILL GATES
First you'll lose your job. Then it gets scary.



STUART RUSSELL
Earth for the earthlings!



NICK BOSTROM
Prepare for "Disneyland without children."



MAX TEGMARK
Uh, can we talk about this?



DEMIS HASSABIS
Full speed ahead!



PETER THIEL
Will be a winner either way.



STEVE WOZNIAK
Resigned to being a robot's pet.



SAM ALTMAN
Sees intergalactic domination—or extinction.



ELON MUSK
Eyeing the next flight to Mars.



LARRY PAGE
Green-lighted Google Brain. 'Nuff said.



YANN LeCUN
Chill, people! We got this.



ANDREW NG
Trust the robot.



MARK ZUCKERBERG
Worried? Tell my A.I. butler.



HIT THE GAS

RAY KURZWEIL
Eager to be a cyborg.

“Accelerationist” vs. “Safetyist”

- Have to outcompete other powers (Russia, China)
- AI development speed-up lead to economic growth, scientific discovery, poverty reduction, higher education, etc.
- “move fast, break things”

- Prioritize minimizing AI risk
- We could make mistakes that spiral out of control!
- “EA”-aligned
- Need to build robust safe-guards first

- AI/AGI could be greatest achievement ever or the biggest existential threat
- AI will self-improvement on a feedback loop, recursively

Bostrom’s Paperclip Experiment!

- Bostrom’s 2003 paper "Ethical Issues in Advanced Artificial Intelligence."
- Introduces the concept of AI Alignment (misalignment)

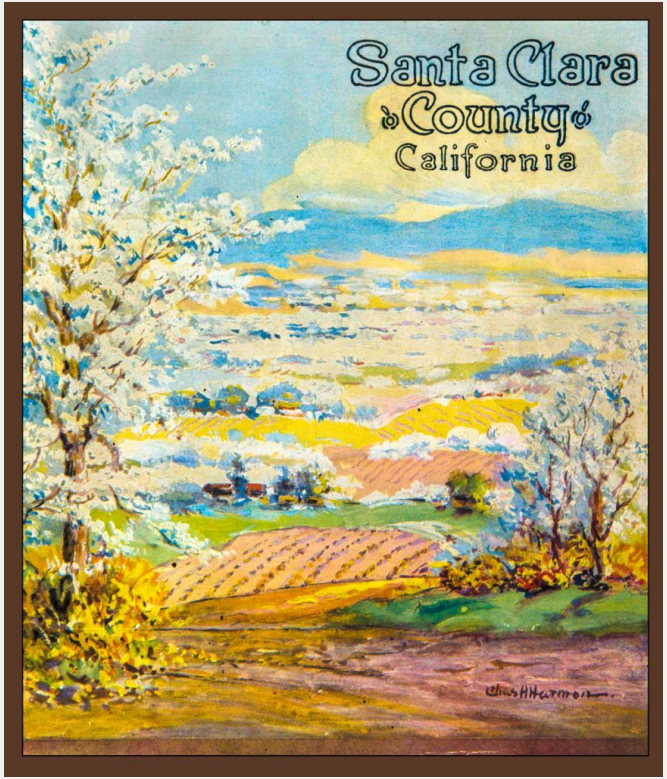


Maximize Paperclip Production!



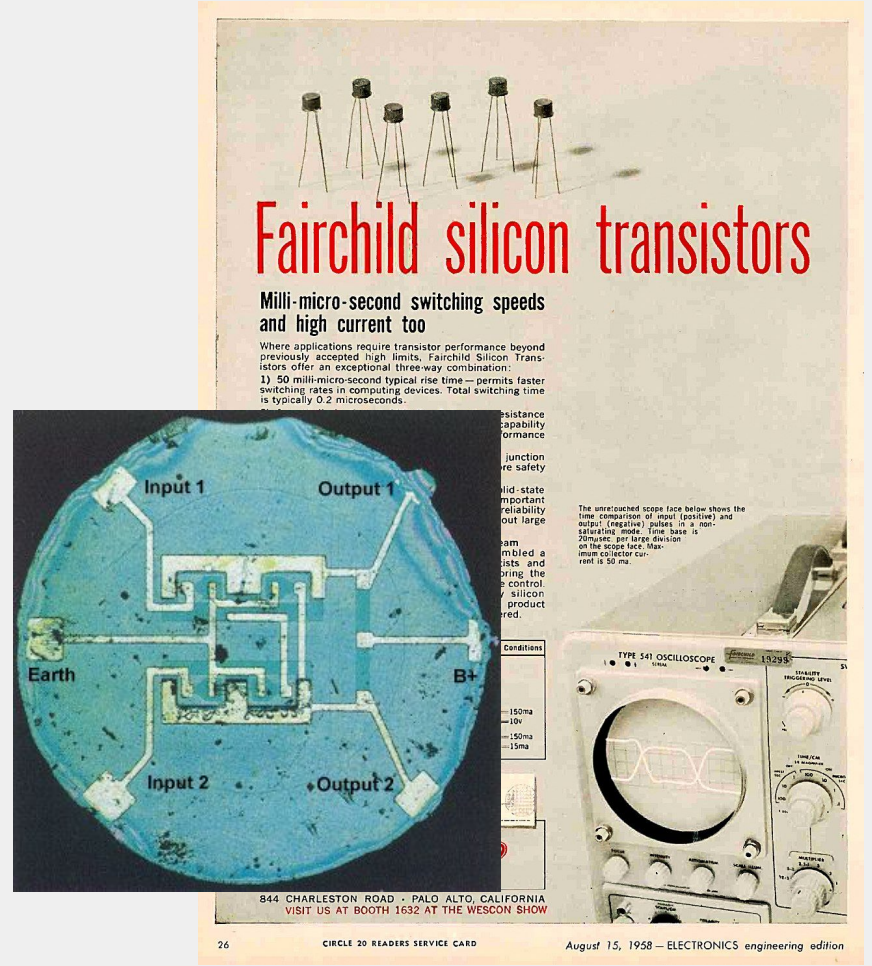
Silicon Valley before Silicon

- “Valley of Heart’s Delight”
- Mostly farmland with 40 canneries
- International prune export
- Delmonte, Sunsweet started here!
- Chinese, Japanese immigrants first, then Italian, East/Southern Europeans
- WWII: gradually replaced by Mexican immigrants (braceros)



Semiconductors

- In 1947 at Bell Labs, John Bardeen, Walter Brattain, and William Shockley invent transistors – can replace bulky vacuum tubes!
- Shockley Semiconductor 1956
- Fairchild Semiconductor 1957 by the “Traitorous Eight”
- IBM bought 2N697 for $150 apiece in order to build the computer for the B-70 bomber.
- Fairchild made the first silicon integrated circuit
- Fairchild spawns 60+ of spinoff semiconductor companies —e.g. Intel (1968, founded by Noyce and Moore)



SRI's SIRI (2000-2008)

- DARPA funds a project called CALO (Cognitive Assistant that Learns and Organizes) —> an assistant that can send emails helping users organize their tasks, etc.
- spin-off of (Personal Assistant that Learns) framework from DARPA, a defense assistant tool
- February 2010, Siri was released as an app for iOS – two months later Apple acquired it and integrated Siri software into its hardware devices.



SIRI 2010

Deepmind 2010

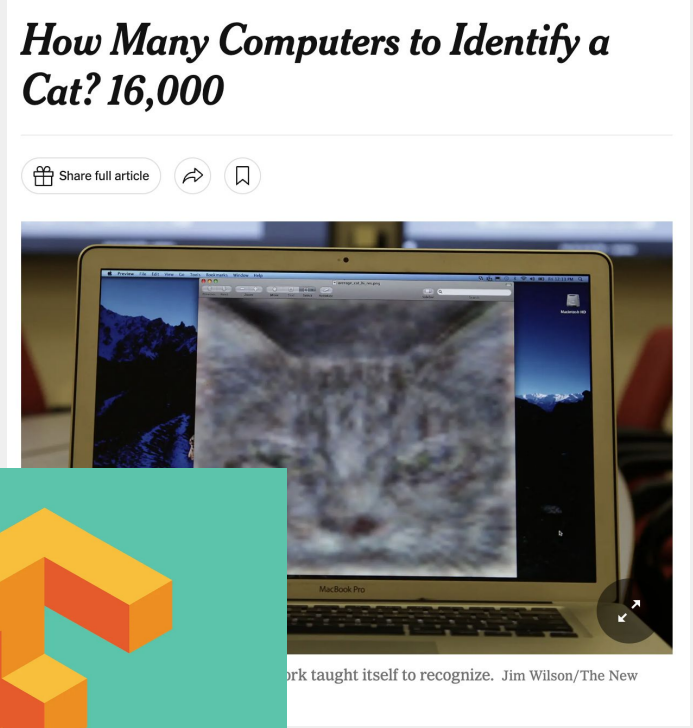
- How to make AGI for the benefit of humanity
- “Could we create a machine that could work backwards to make sense of the universe? AGI would give you an insight into where we came from, and what is God.”
- Demis Hassabis, Shane Legg and Mustafa Suleyman cofounders
- Early investment from Peter Thiel, Elon Musk, Jaan Thallin
- Met investors at Singularity Summit
- Earliest projects on reinforcement learning with game screen pixels as input
- Ethics board





Google Brain 2011

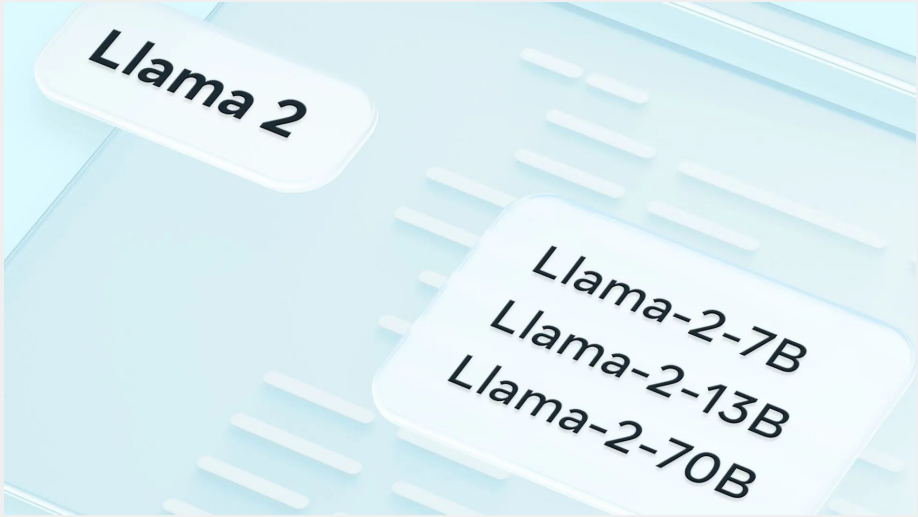
- Unsupervised “cat experiment” 2012: Stanford professor Andrew Ng. One of the group’s first projects was training a cluster of 16,000 processors in 1,0000 machines on neural nets to recognize a cat based in 10 million images from Youtube videos.
- Focus on Google Ads
- Customer AI Chips (TPUs 2016)
- 2015 Tensorflow 1.0





Facebook FAIR 2013

- Yann LeCunn heads FAIR and pushes open source
- PyTorch 2016
- RoBERTA 2019
- Llama series



PyTorch

OpenAI 2015

- Sam Altman et al.
- Some handful of Deepmind researchers
- Non-profit dedicated to AI safety
- Elon Musk wants to fund Deepmind rival, “Hitler of AI”
- Altman email to Musk on May 25, 2015: “someone other than Google” should build AGI first.



Anthropic 2021

- Delaware Public Benefit Corporation
- Dario Amodei & Daniela Amodei from AI Safety team OpenAI
- 7 former OpenAI employees
- Funds from Google, Amazon
- FTX founders support out of effective altruism



ChatGPT 2022

- Just the decoder part of Transformers?
- GPT, GPT2, GPT3, DALL-E 2
  - Gradually less transparent!
- After two months online ChatGPT had attracted thirty million registered users, (one of the fastest-growing online services in history)
- By early 2024, around one hundred million people.



Aug 29, 2024 - Technology

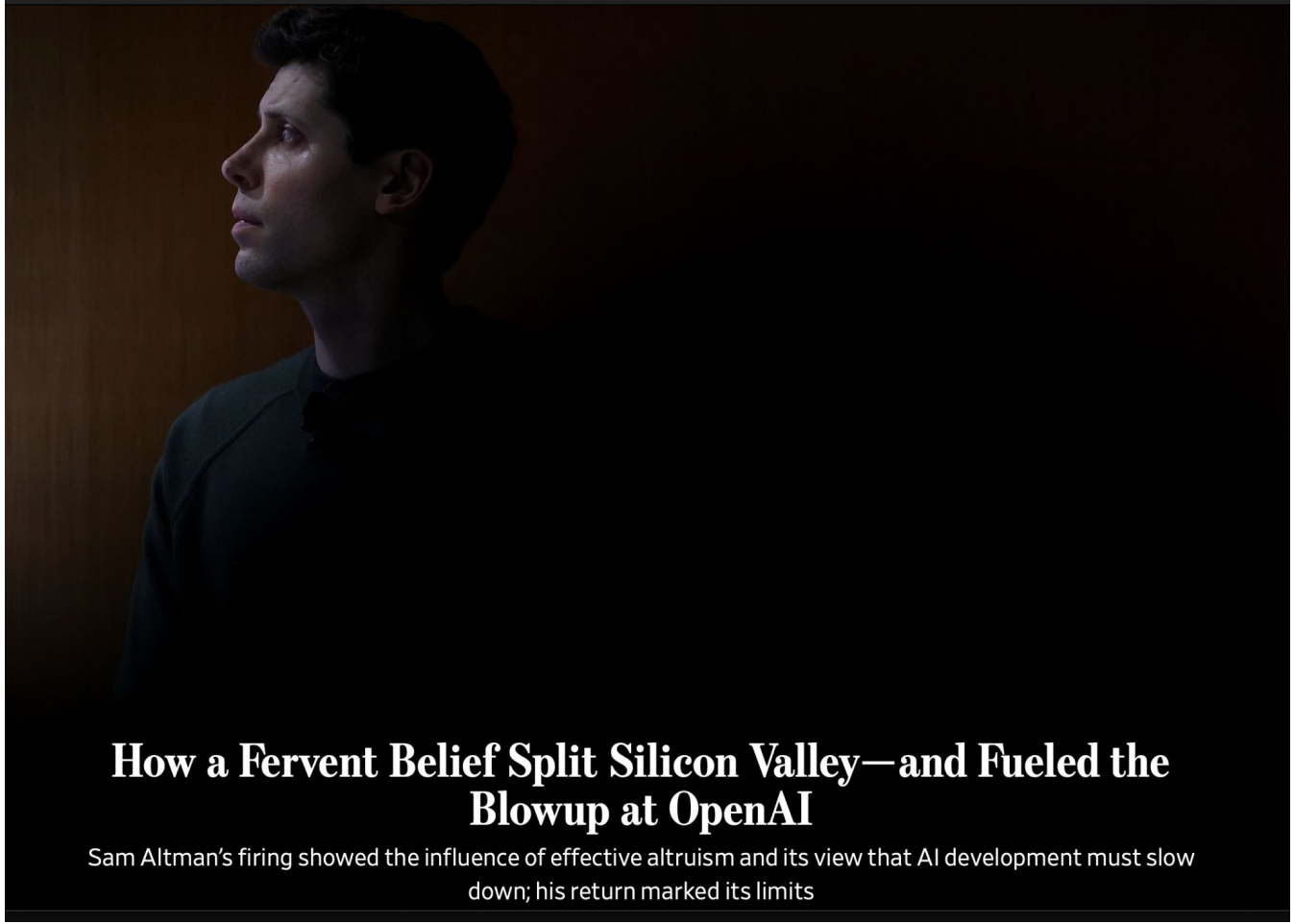
OpenAI says ChatGPT usage has doubled since last year

Ina Fried

OpenAI said on Thursday that ChatGPT now has more than 200 million weekly active users — twice as many as it had last November.

THE WALL STREET JOURNAL.

SIGN IN



How a Fervent Belief Split Silicon Valley—and Fueled the Blowup at OpenAI

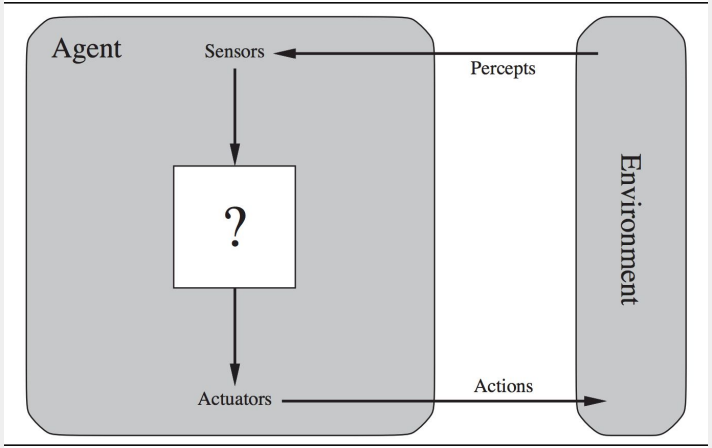
Sam Altman's firing showed the influence of effective altruism and its view that AI development must slow down; his return marked its limits

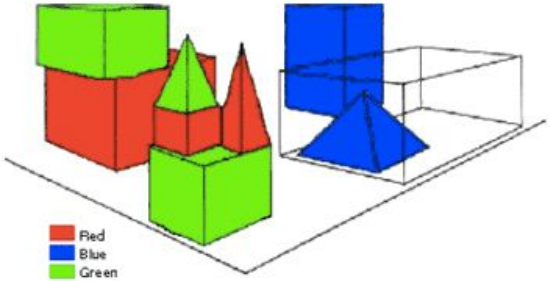
This week!

- Are games and AI a match made in heaven?
- Similarities & differences in AI for health and AI for games
- What are different factors at play in how different domains adopt AI?

Games, a natural AI habitat? Microworlds 1972

- A limited world with defined rules, perfect place to build microworlds
- Natural confined environment to test AI





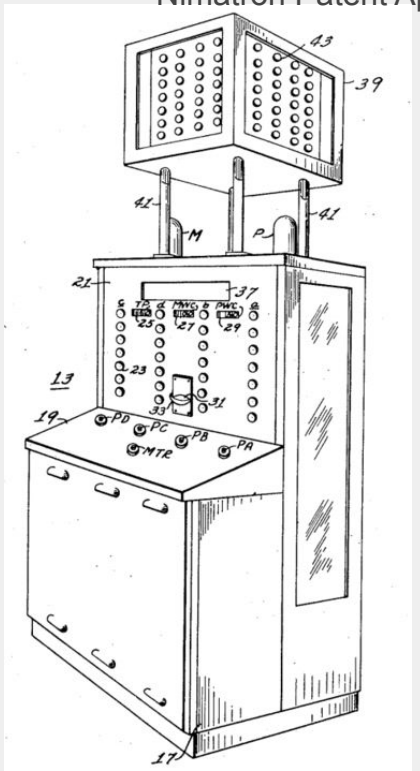
Person: Pick up a big red block.
Computer: OK.
Person: Grasp the pyramid.
Computer: I don't understand which pyramid you mean.

Nim on Nimatron 1940



Nimatron, New York World Fair, 1940
(Westinghouse Corp.)

Nimatron Patent Application

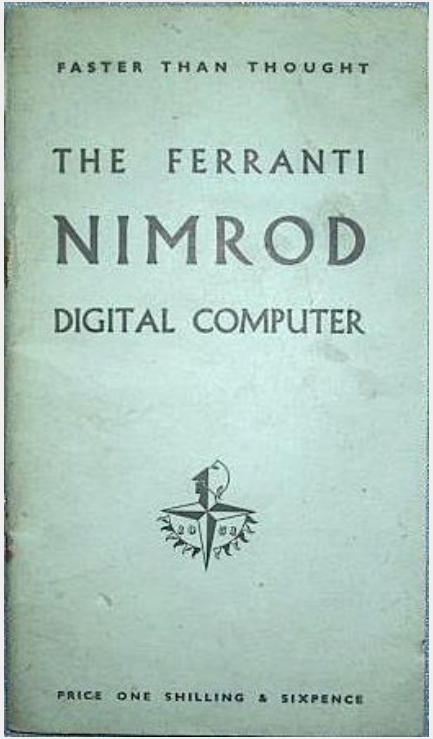


- For the NY World Fair
- played 100,000 times
- If won, NIM CHAMP coin!
- World's first arcade game?
- Nimatron won 90,000 times

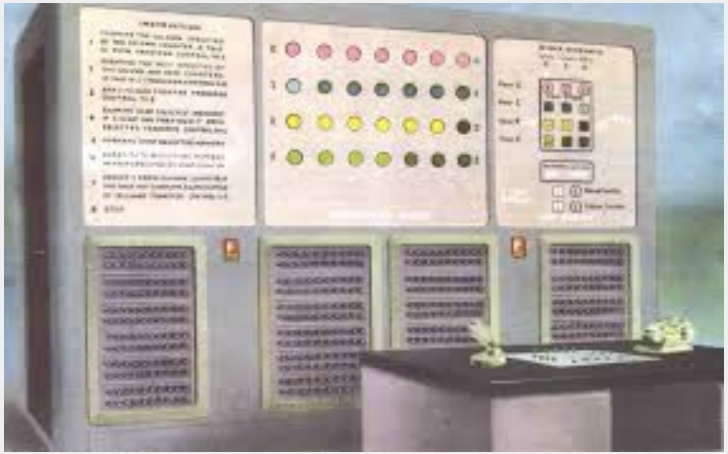


Nim on Nimrod 1951





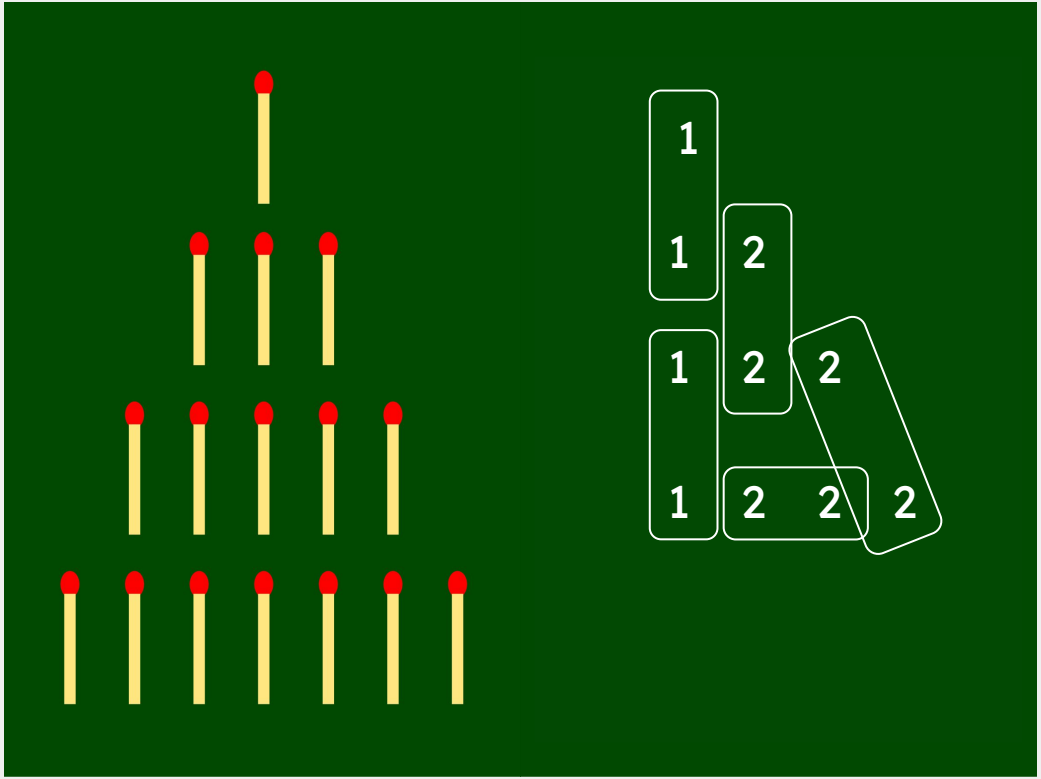
- Alan Turing managed to beat the computer!
- “COMPUTER WINS”



Nimrod, The Festival of Britain, 1951 (Ferranti)

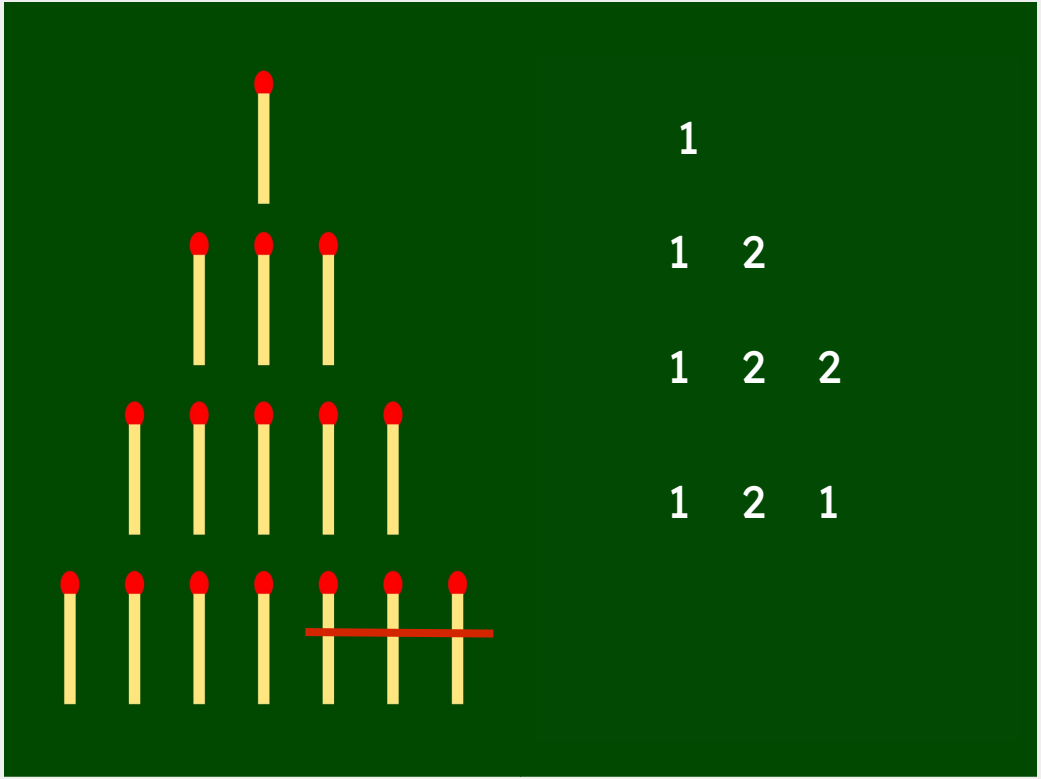
Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent



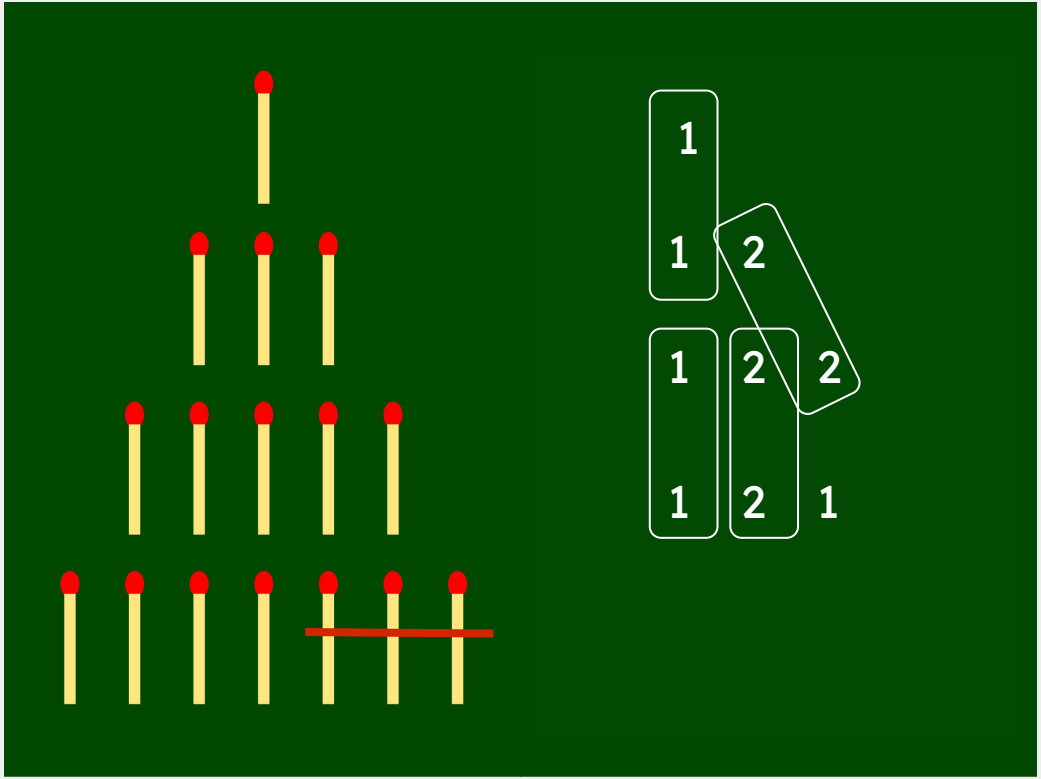
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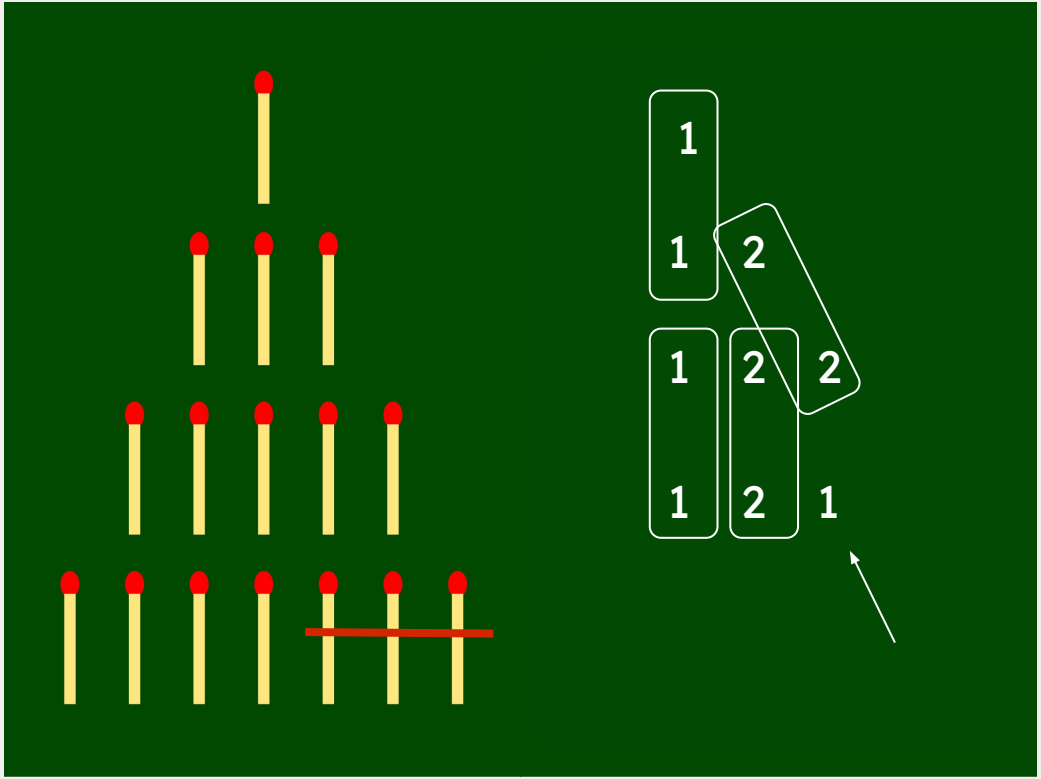
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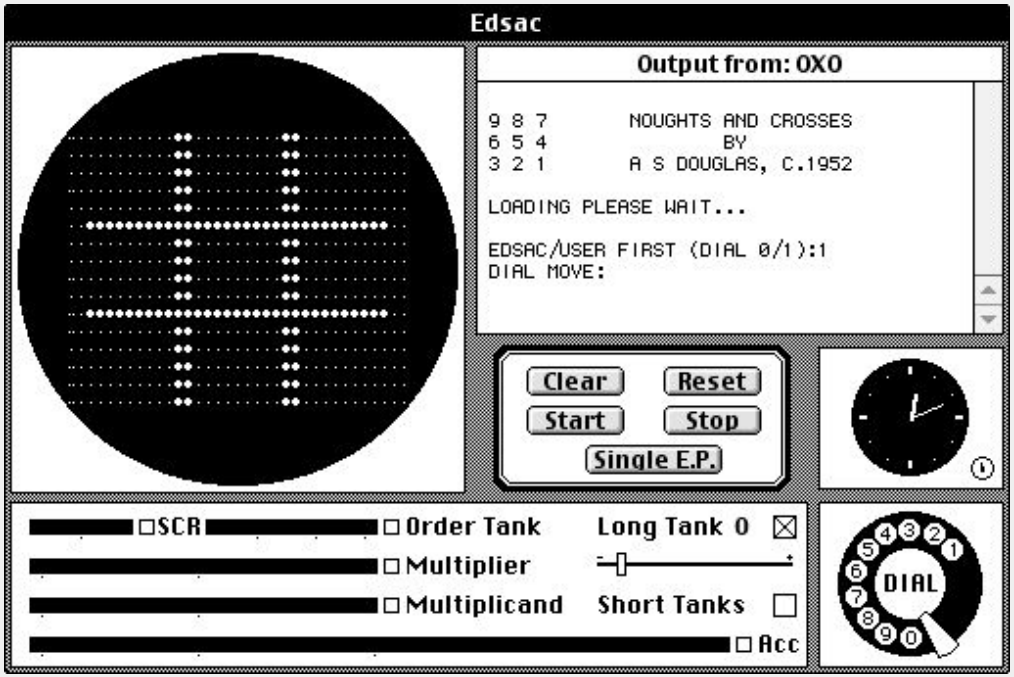


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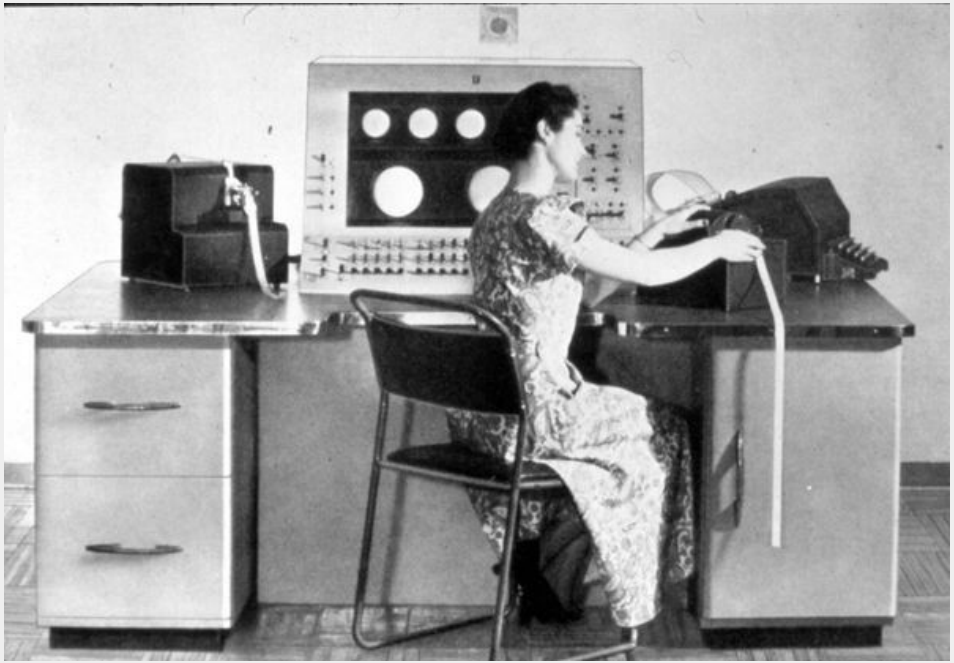
OXO: AKA Tic-Tac-Toe 1951



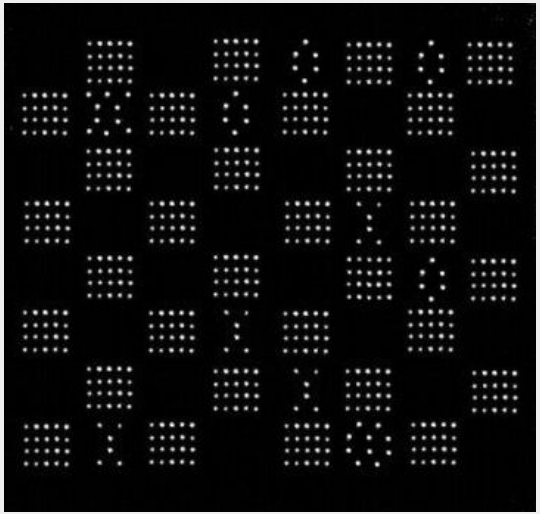
- AI opponent for Tic-Tac-Toe → minimal tie when played optimally
- On the EDSAC, early british computer at Cambridge
- One of first computer games on a screen (not light bulbs)

Checkers on Ferranti Mark I 1952

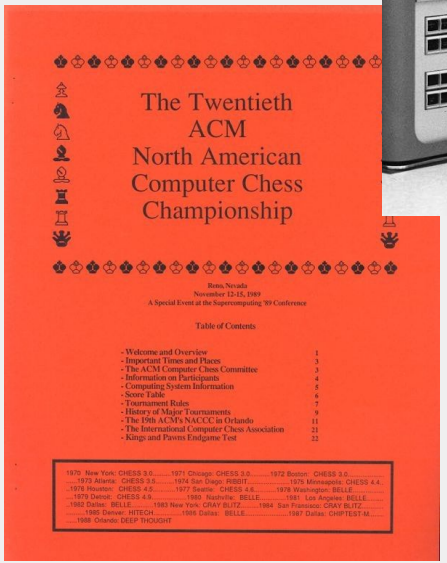
- On the Ferranti Mark I, world's first commercially available digital computer (4,050 vaccuum tubes)
- "Draughts" first ever played July 1952



Ferranti Mark I 1951



Chess programming begins 1957



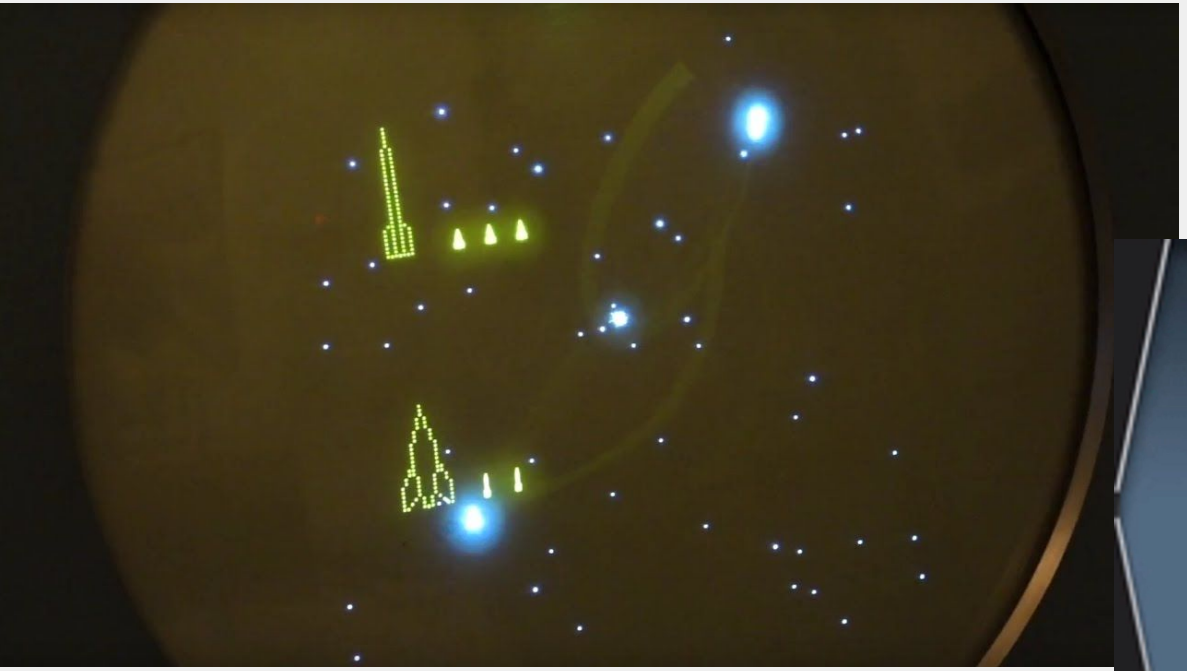




- 1957: First computer chess game IBM 704; evaluate 42,000 instructions per second
- 1962: IBM 7040; 1,100 positions per second
- 1980s: Feng-hsiung Hsu writes ChipTest that goes on to win the North American Computer Chess Championship in 1987→ later helps develop DeepBlue

Spacewars! 1962

https://www.youtube.com/watch?v=AbAODTne64w



- MIT students made Spacewars! on PDP-1 (first mini computer)
- Widely shared and inspired Atari founders
- Rudimentary AI as second player



Pong by Atari 1972

https://www.youtube.com/watch?v=fhd7FfGCdCo



- The “opponent” AI paddle programmed to follow the ball



Deep Blue 1997



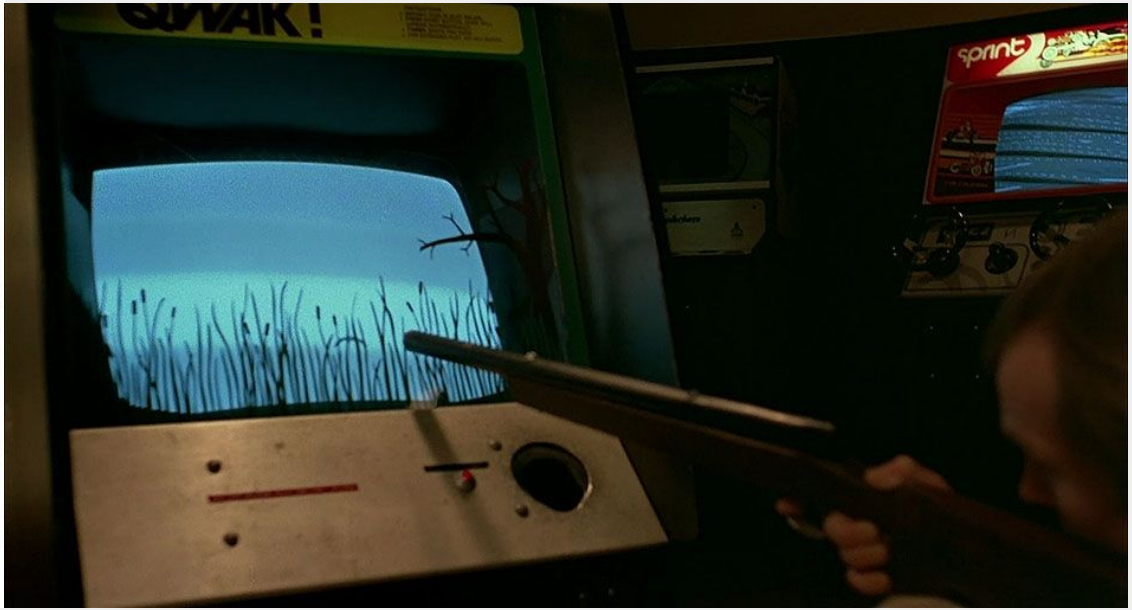
- IBM’s Deep Blue beats Garry Kasparov, previously undefeated chessmaster, in chess
- 6 matches (Deep Blue won first game) → 4:2 Kasparov wins
- Evaluate 200 million chess positions per second



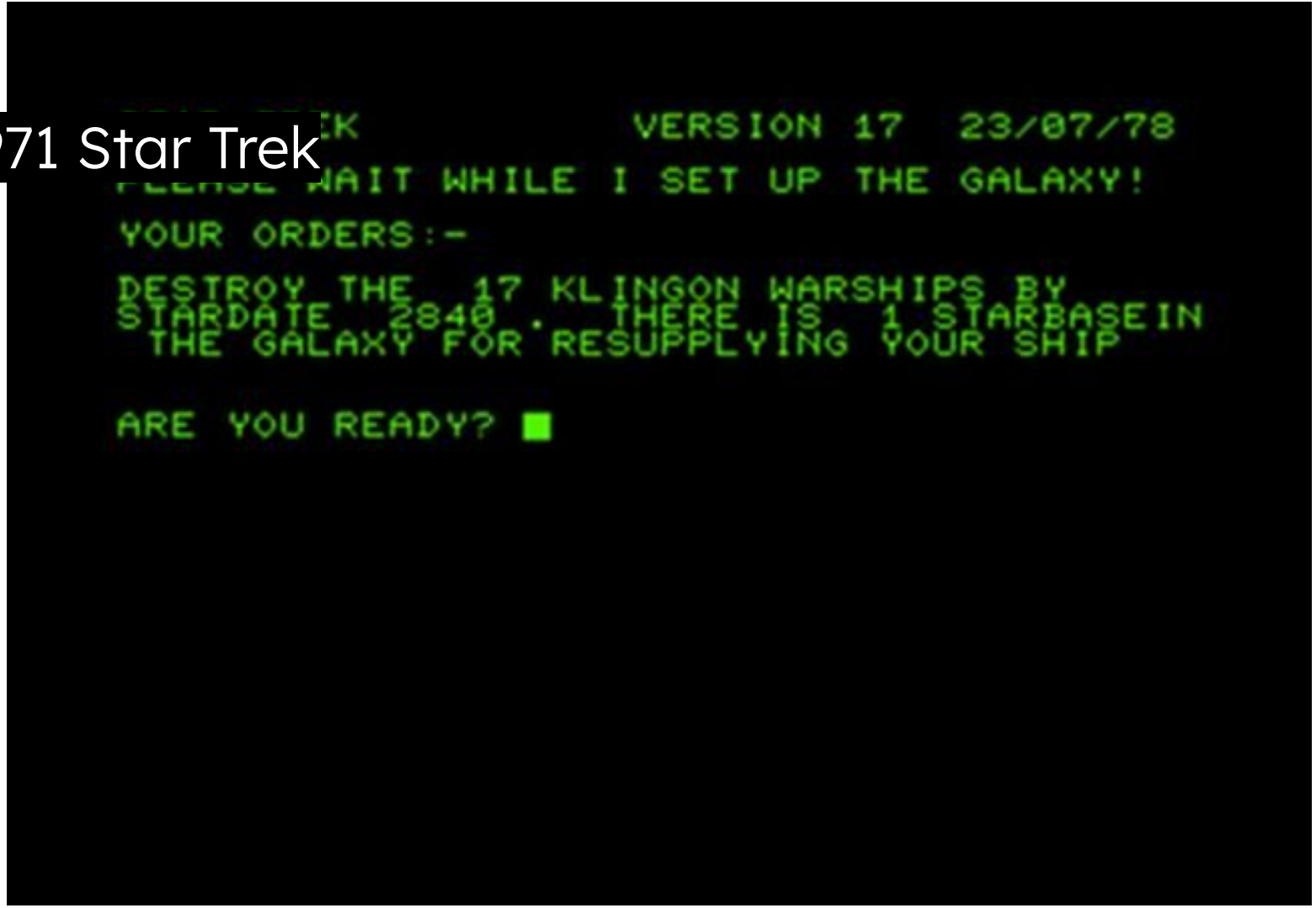
AI as NPCs 1970s



- Arcade Golden Era 1970s
- Atari's QWAK 1973 first AI NPC



1971 Star Trek



SPACE INVADERS 1978

https://www.youtube.com/watch?v=viDiuf9Mv7o



- Wildly popular game
- Lots of diversity in computer opponent behavior
- Increasing difficulty levels, distinct movement patterns, and in-game events (that hashed player input).



1980

- First brought the maze to gaming AI

Real-time Strategy (RTS) Games 1990s



Herzog Zwei 1989

https://www.youtube.com/watch?v=BYfYrhT82RM

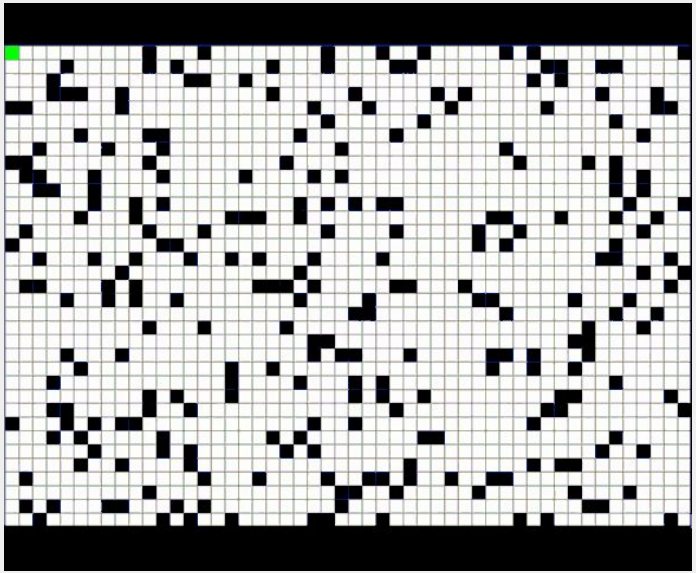
- Real-time strategy games pose new challenge for game makers
- Mostly scripted, fixed opponent responses

Real-time Strategy Games 1990s

- Decisions trees for rule-based AI
- Pathfinding for NPCs in map navigation (Age of Empires, Civilization)



Age of Empires

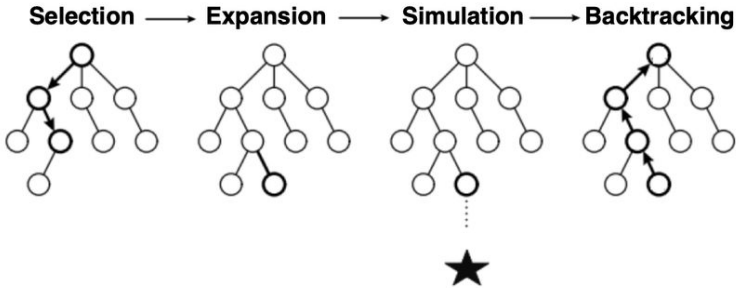


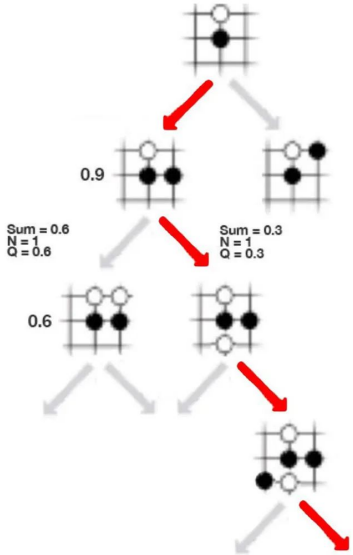
A* pathfinding

Deep Learning & Reinforcement Learning 2010s

- In the mid 2010s, deep learning was used to successfully beat international masters in Go such as AlphaGo.
- Monte Carlo Tree Search methods for games like Chess & Shogi

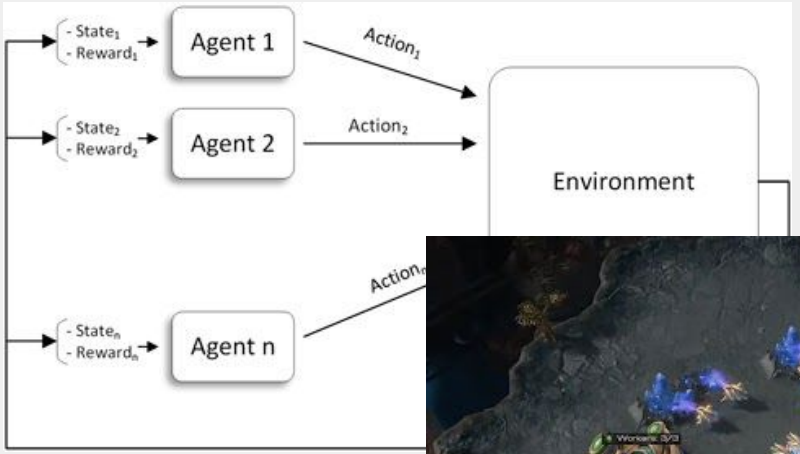






Deepmind's AlphaStar 2019

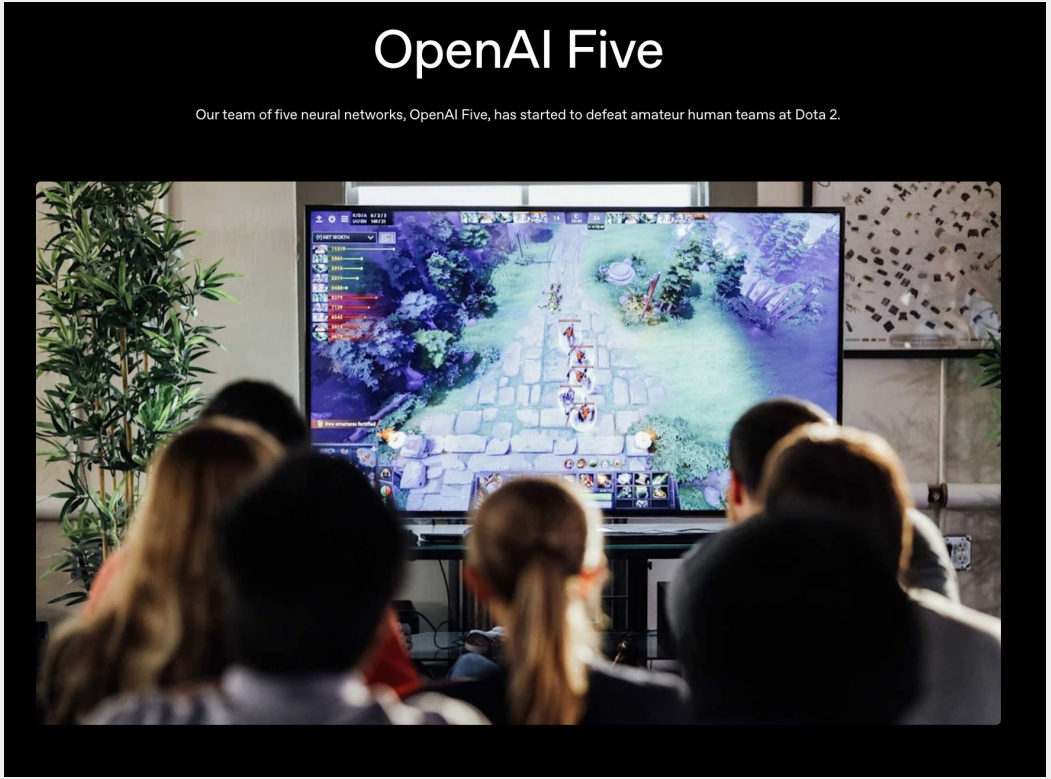
https://www.youtube.com/watch?v=6EQAsrfUIyo





- Late 2010s people began to experiment with multi-agent reinforcement learning methods to play complex RTS games like Starcraft II
- Dec 2018, AlphaStar beats world class Starcraft player Team Liquid's Grzegorz "MaNa" Komincz 5-0

OpenAI's Five masters Dota 2 2019

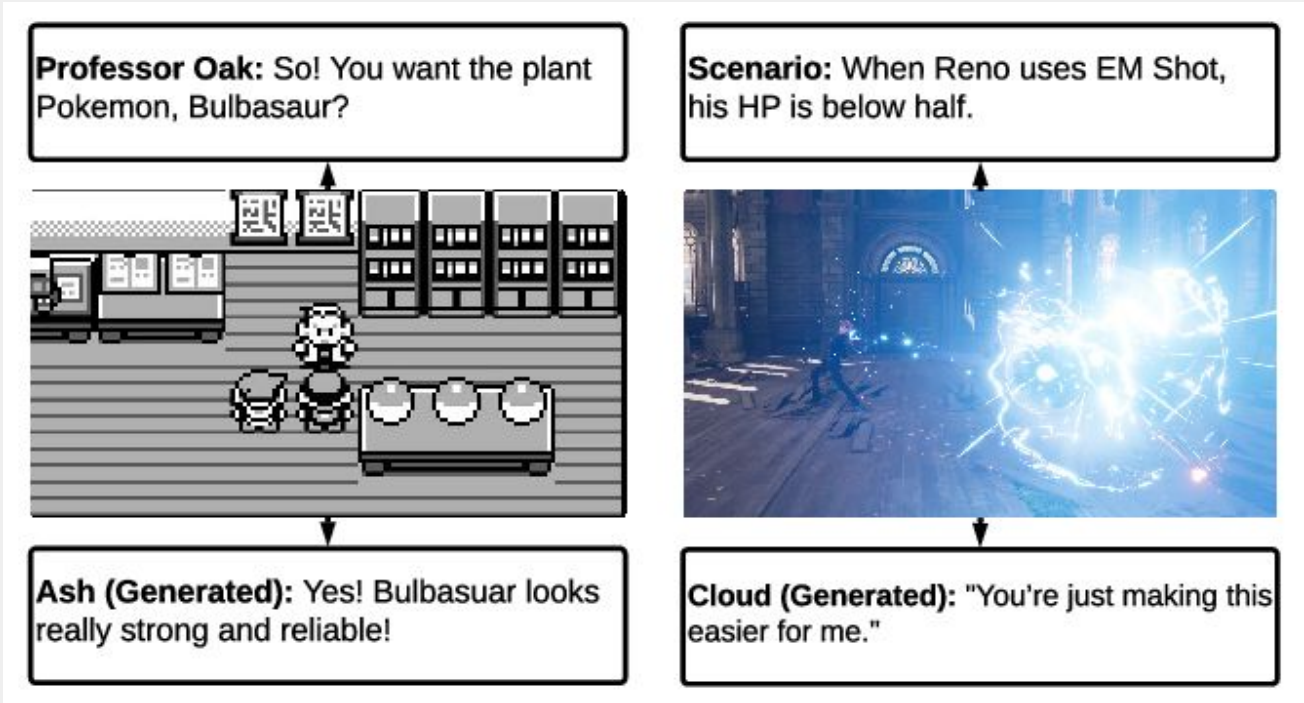


- April 2019, OpenAI Five wins 2 out of 3 games against human e-sports team OG
- Played 45,000 years of Dota2 (self-play)



AI for Game Content Generation

Dialogue Generation for NPCs or story games



Explore Playground Beta Pricing Docs Blog Changelog Sign in Get started

lambdal / text-to-pokemon

Generate Pokémon from a text description

Cold Public 7.9M runs T4 GitHub License

https://replicate.com/lambdal/text-to-pokemon/examples

Run with an API

Playground API Examples README Versions

Prediction

lambdal/text-to-pokemon:ff6cc781

Model	ID	Status	Source	Hardware	Total duration	Created
lambdal/text-to-pokemon:3554d9e6	n2vfz327yvgnpnrcbecukyqvh4	Succeeded	Web	–	30.2s	over 2 years ago by @justinpinkney

Input

Form JSON Node.js Python HTTP Cog Docker

prompt

```
Yoda
```

num_outputs

```
1
```

guidance_scale

```
7.5
```

num_inference_steps

```
"50"
```

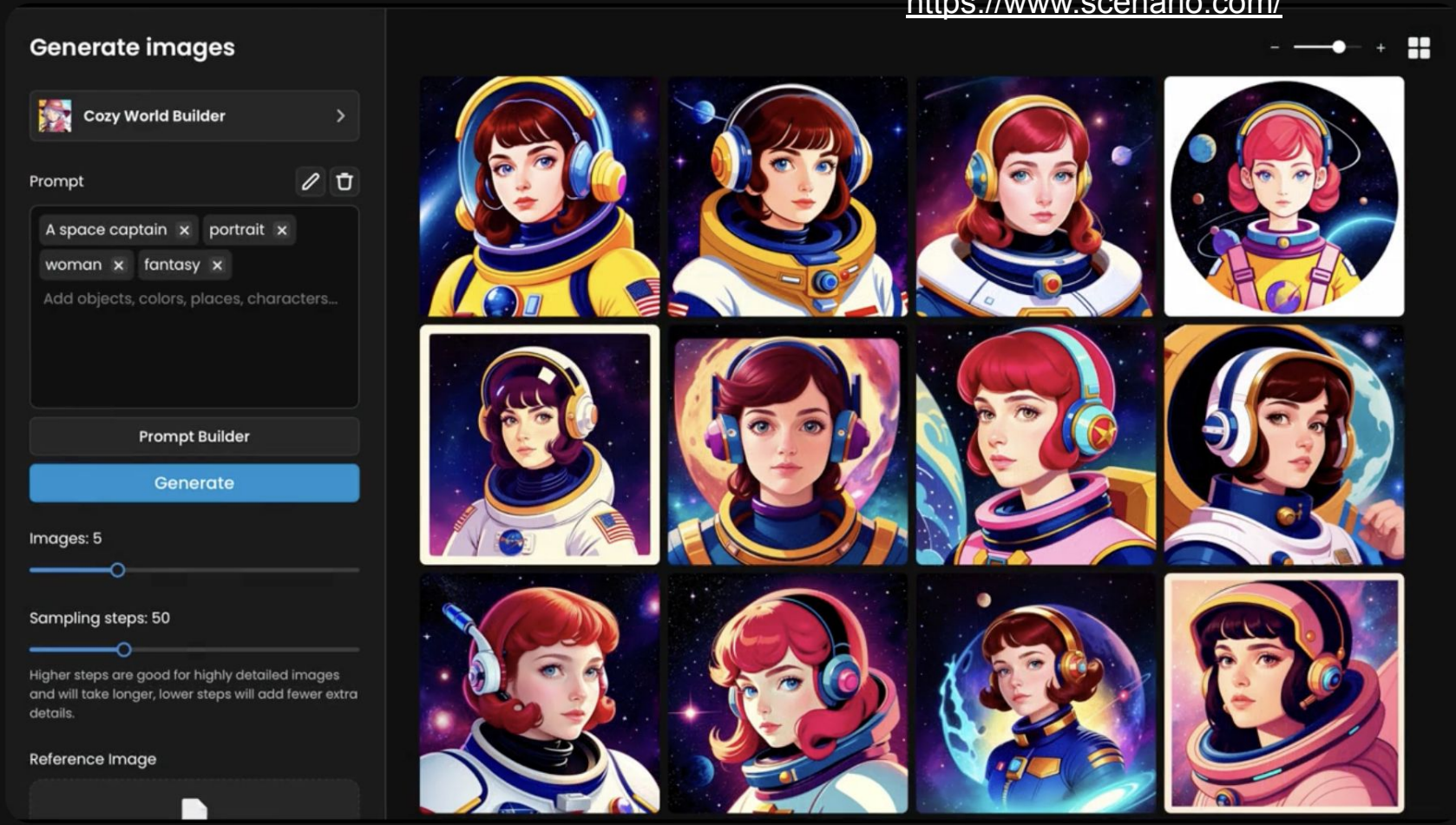
Output

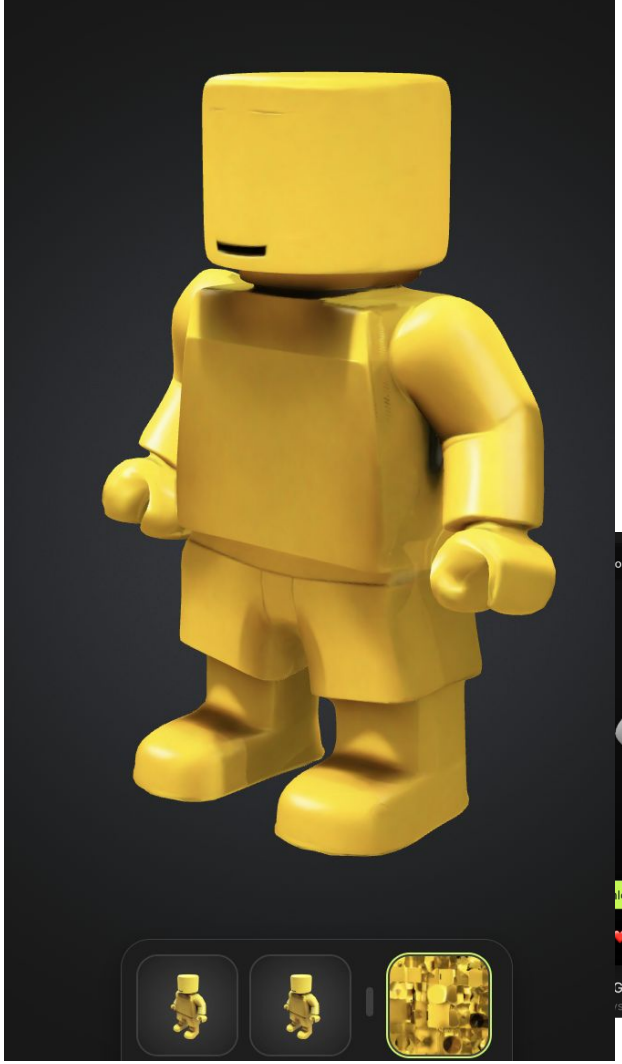
Preview JSON



ps://replicate.com/lambdal/text-to-pokemon/versions

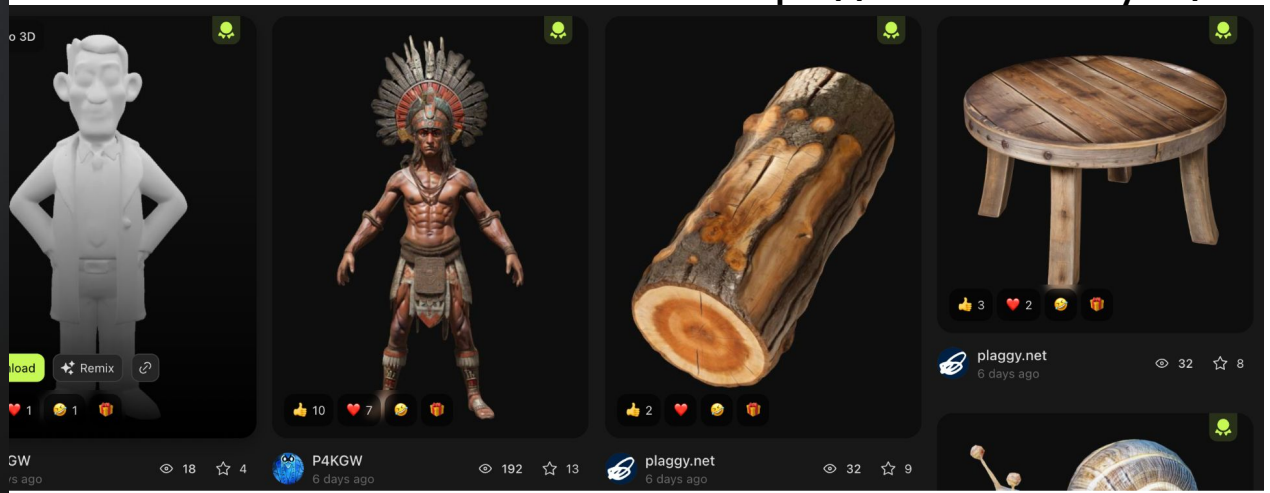
https://www.scenario.com/





Sound effects, backgrounds, voice, 3d mesh, etc.

https://www.meshy.ai/

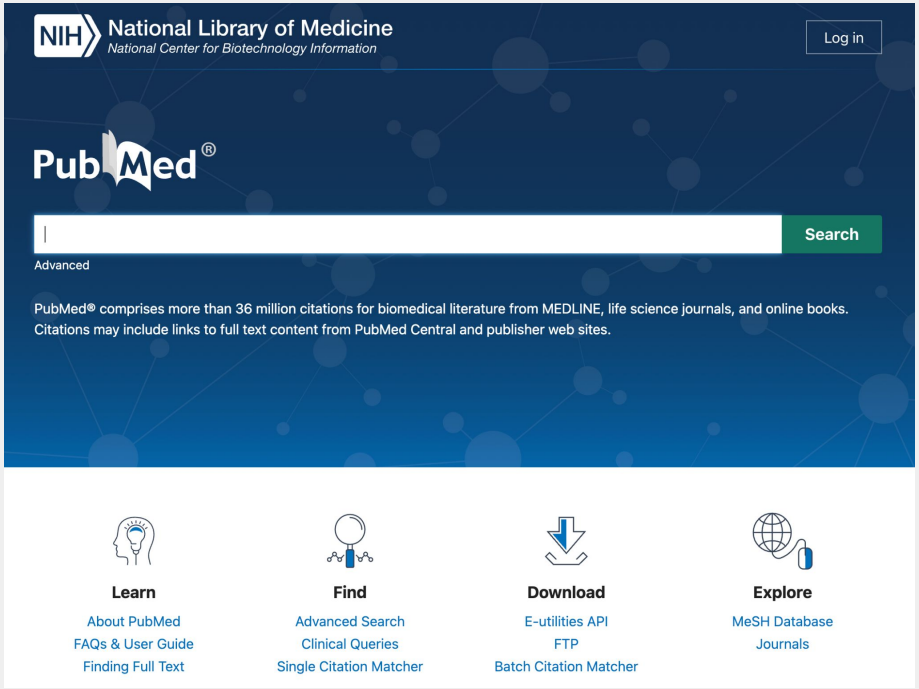


AI in Health

- diagnosis of diseases
- prediction of therapeutic responses
- preventative medicine
- improving provider workflow & clinical operations
- disease and therapy monitoring
- improving procedural accuracy and overall patient outcomes

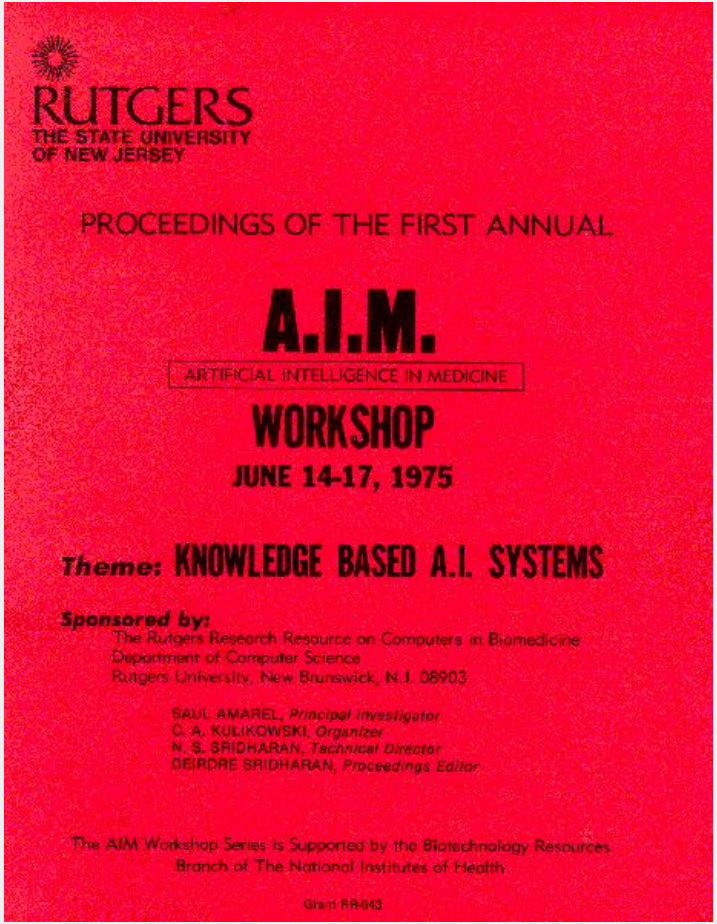
Digitizing Health Data 1950s

- While the AI field was moving along with experiments with robots (Shakey) and chatbot (ELIZA), medicine was slower to respond to automation
- The medical field began with digitization of health data
- Medical Literature and Analysis and Retrieval System MEDLARS; online counterpart: MEDLINE, 30 million references to journal articles in life sciences with a concentration on biomedicine, 1789 to present)
- and search engines of medical records (PubMed by National Library of Medicine)



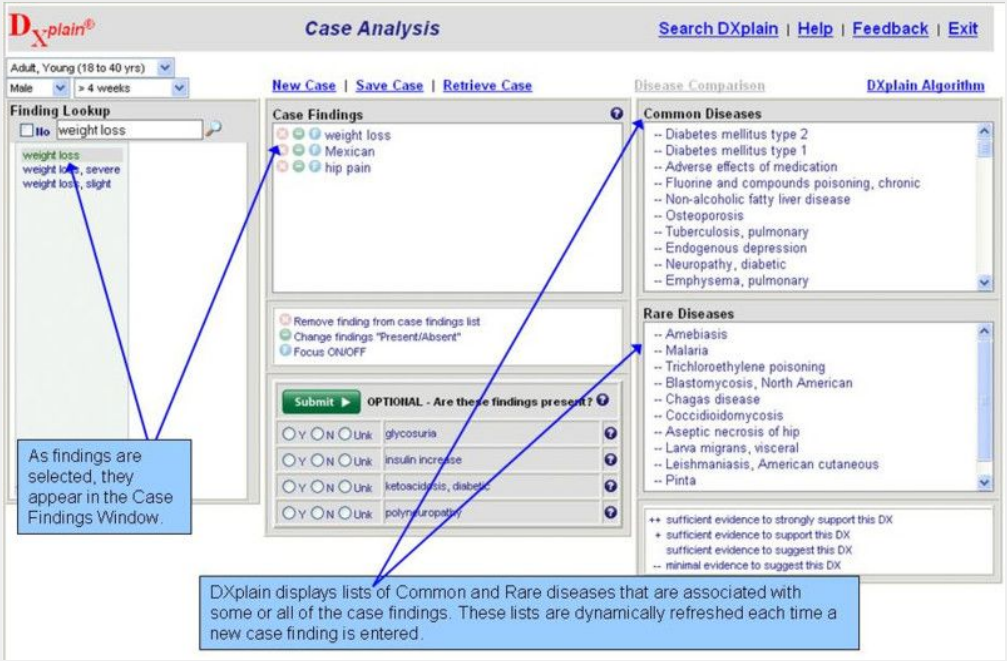
First AI Conferences 1970s

- Researchers in medicine worked on information sharing and networking systems for medicine.
- In 1971, The Research Resource on Computers in Biomedicine by Saul Amarel at Rutgers University. The Stanford University Medical Experimental – AI in Medicine created in 1973.
- The first AI in medicine workshop was held in 1975 at Rutgers University hotels by the National Institute of Health.



Medical Expert Systems 1970s

- In the 1970s these groups worked on expert system prototypes for applying AI to medical consulting.
- 1976 CASNET: model-building, consultation, database all brought together to apply information about diseases to individual patients and give physicians advice on patient management.
- 1970s MYCIN: backward chaining AI system with about 600 rules. Physicians could input patient information to get back list of potential bacterial pathogens and antibiotic recommendations.
- 1986 DXplain: input symptoms to get back different diagnoses. Served as an electronic medical textbook with diseases and references. At first release could handle 500 diseases.



Medical Chatbots 2010s

- Pharmabot developed in 2015 to assist in medication education for pediatric patients.
- Mandy was created in 2017 as an automated patient intake process for a primary care practice.

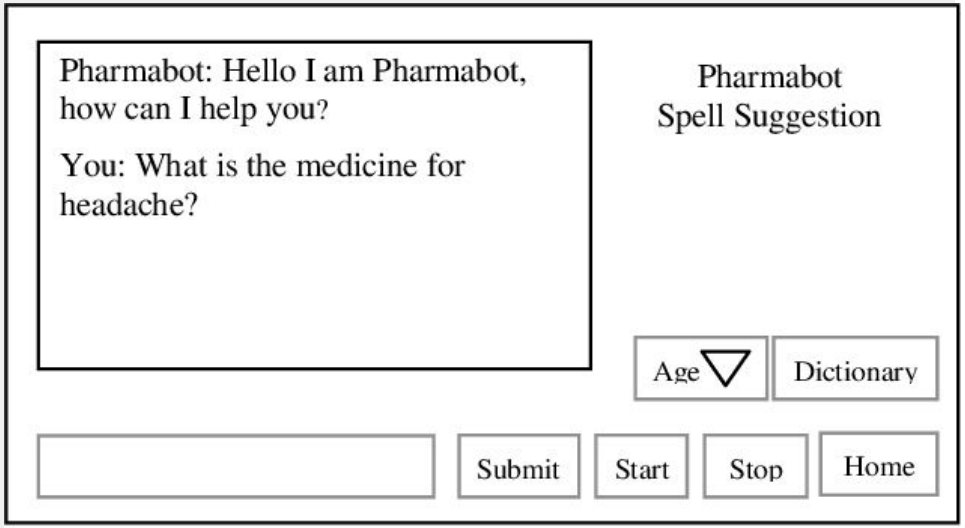


Figure 3. Conversation window

MANDY: Towards A Smart Primary Care Chatbot Application

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Abstract. The paper reports on a proof-of-concept of Mandy, a primary care chatbot system created to assist healthcare staffs by automating the patient intake process. The chatbot interacts with a patient by carrying out an interview, understanding their chief complaints in natural language, and submitting reports to the doctors for further analysis. The system provides a mobile-app front end for the patients, a diagnostic unit, and a doctor's interface for accessing patient records. The diagnostic unit consists of three main modules: An analysis engine for understanding patients symptom descriptions, a symptom-to-cause mapper for reasoning about potential causes, and a question generator for deriving further interview questions. The system combines data-driven natural language processing capability with knowledge-driven diagnostic capability. We evaluate our proof-of-concept on benchmark case studies and compare the system with existing medical chatbots.

Keywords: Medicare chatbot; Patient interview; Natural language processing; AI and healthcare

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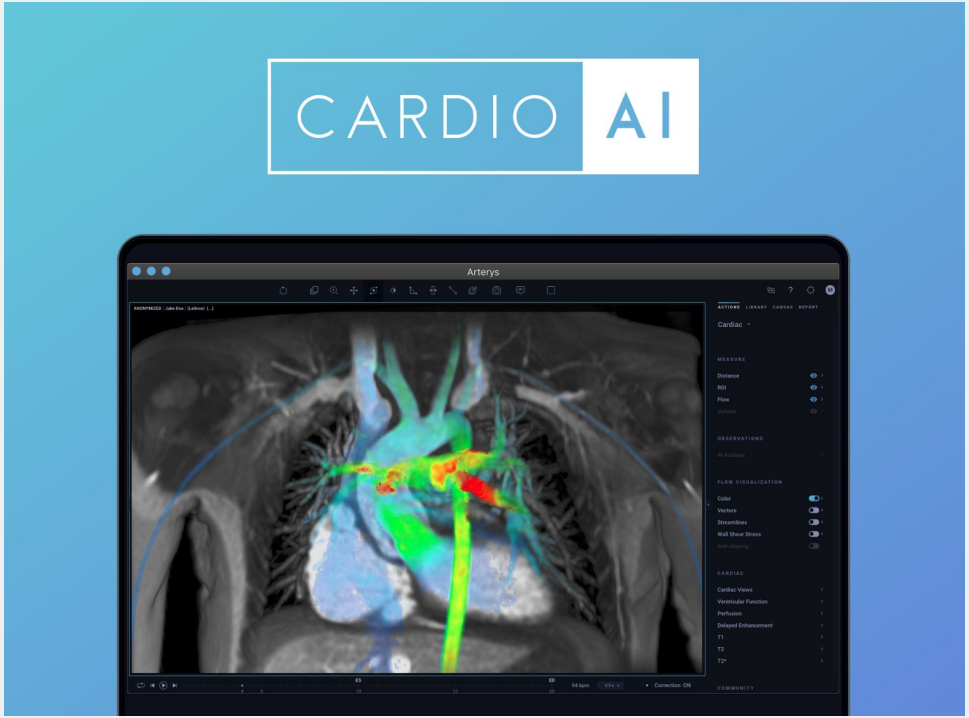
1 Introduction

Patients arriving at a primary care service sometimes need to wait for a long time before being advised by a doctor [26]. This is often due to high workload and limited resources at the primary care service [7]. To facilitate the process, nurses and other health care staffs usually take the role of patient intake. An incoming patient would be first greeted by a receptionist who carries out an intake inquiry. The receptionist would typically be someone who has a certain level of medical proficiency, and the inquiry involves collecting patient information and understanding the symptoms of the patient. A brief report is generated

Deep Learning in Medicine 2010s

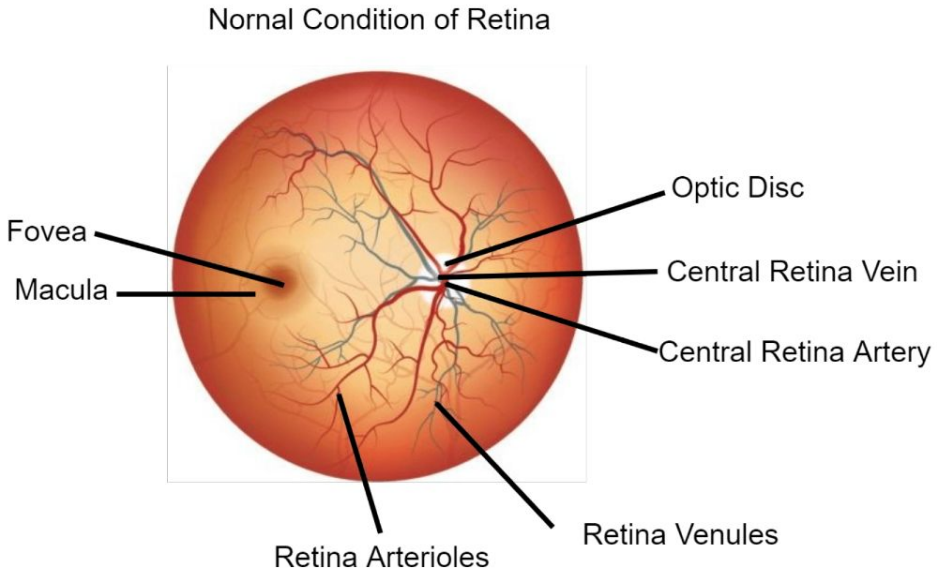
- In the 2010s researchers began to apply deep learning to medical imaging.
- Arterys was the first US FDA approved clinical DL health care product in 2017.
- CardioAI was able to analyze cardiac MRIs in seconds to give information about cardiac ejection fraction (now includes liver and lung imaging, chest and musculoskeletal xray images)

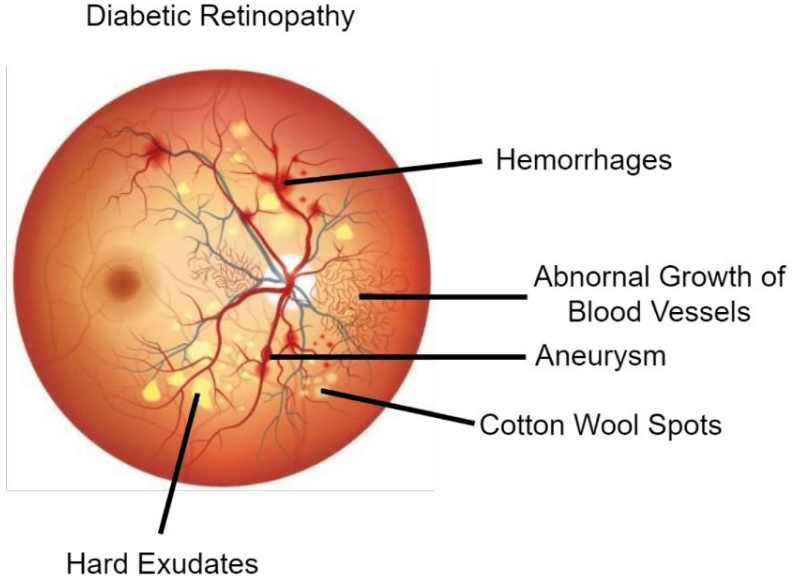
ARTERYS



Deep Learning in Medicine 2010s

- Vision for lesion detection, cardiovascular risk, progression of Alzheimers, and screen for diabetic retinopathy, etc.

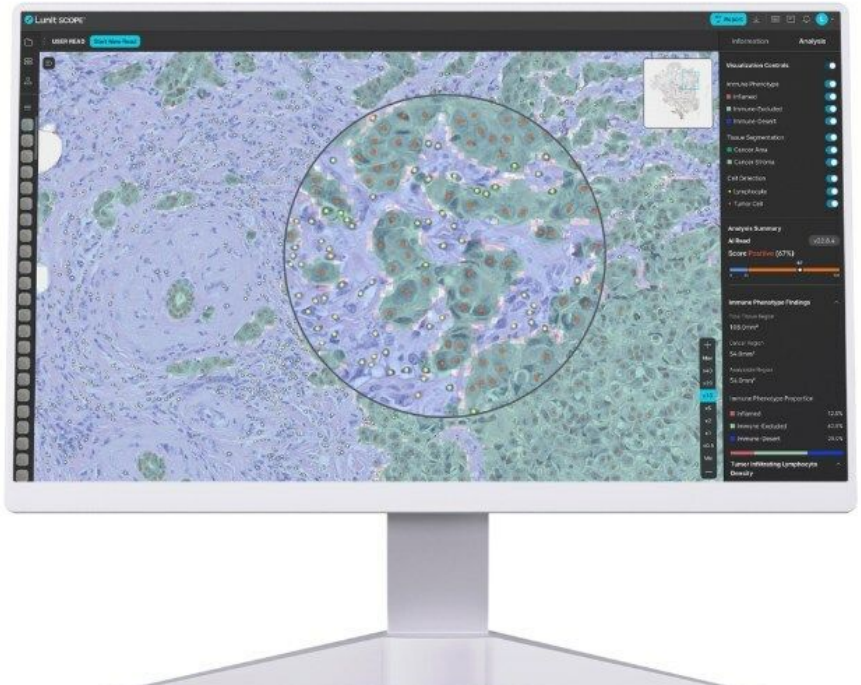


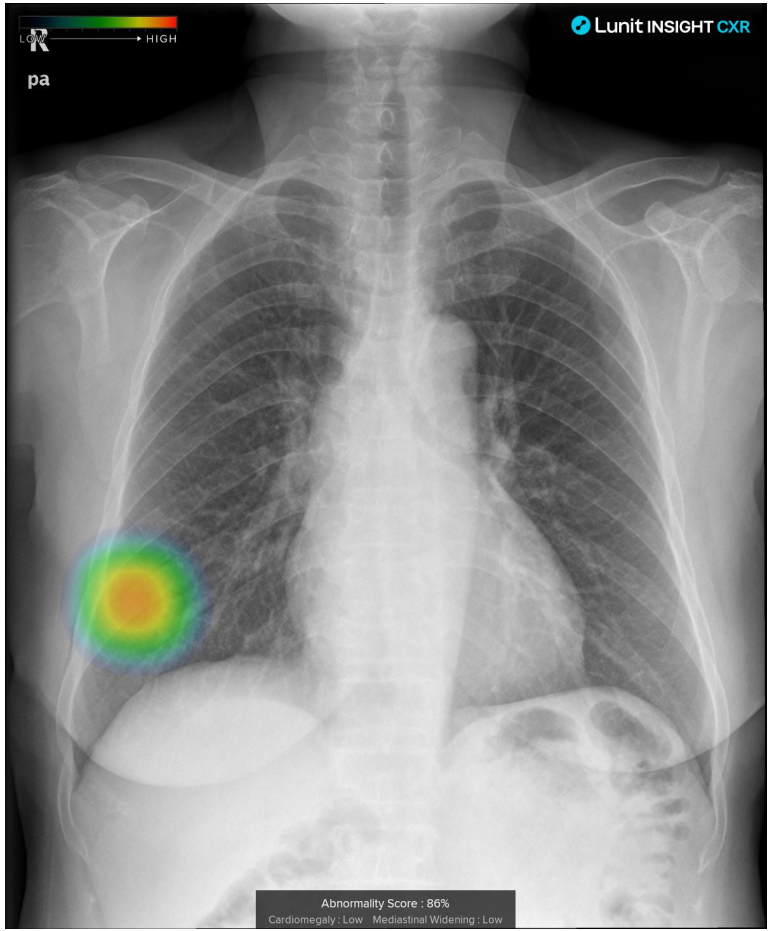


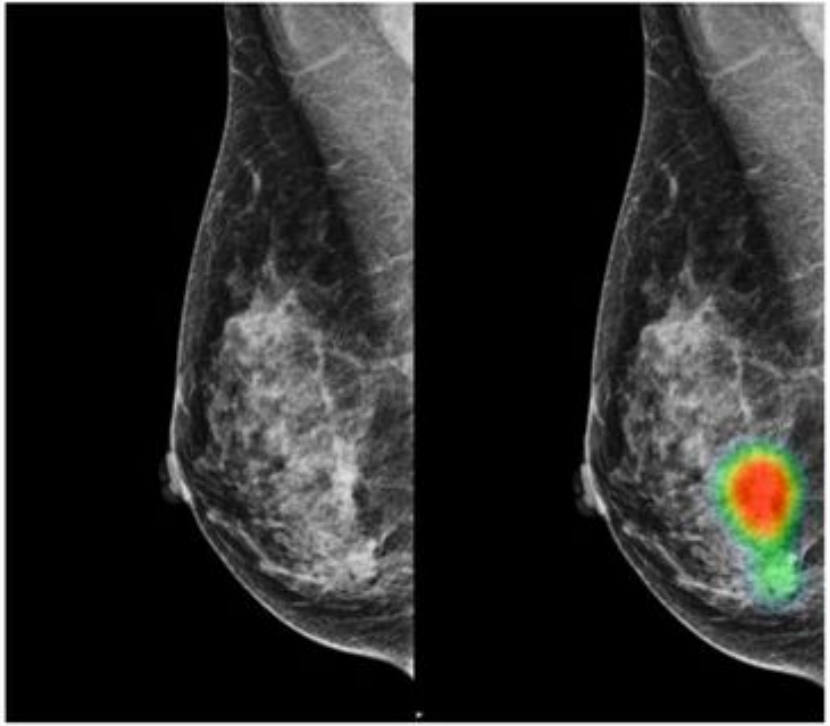
Computer Vision for Cancer Detection 2010s

- Computer vision has recently been used for cancer detection – Lunit is a South Korean company that has been successful with Lung and Breast cancer detection.

Lunit



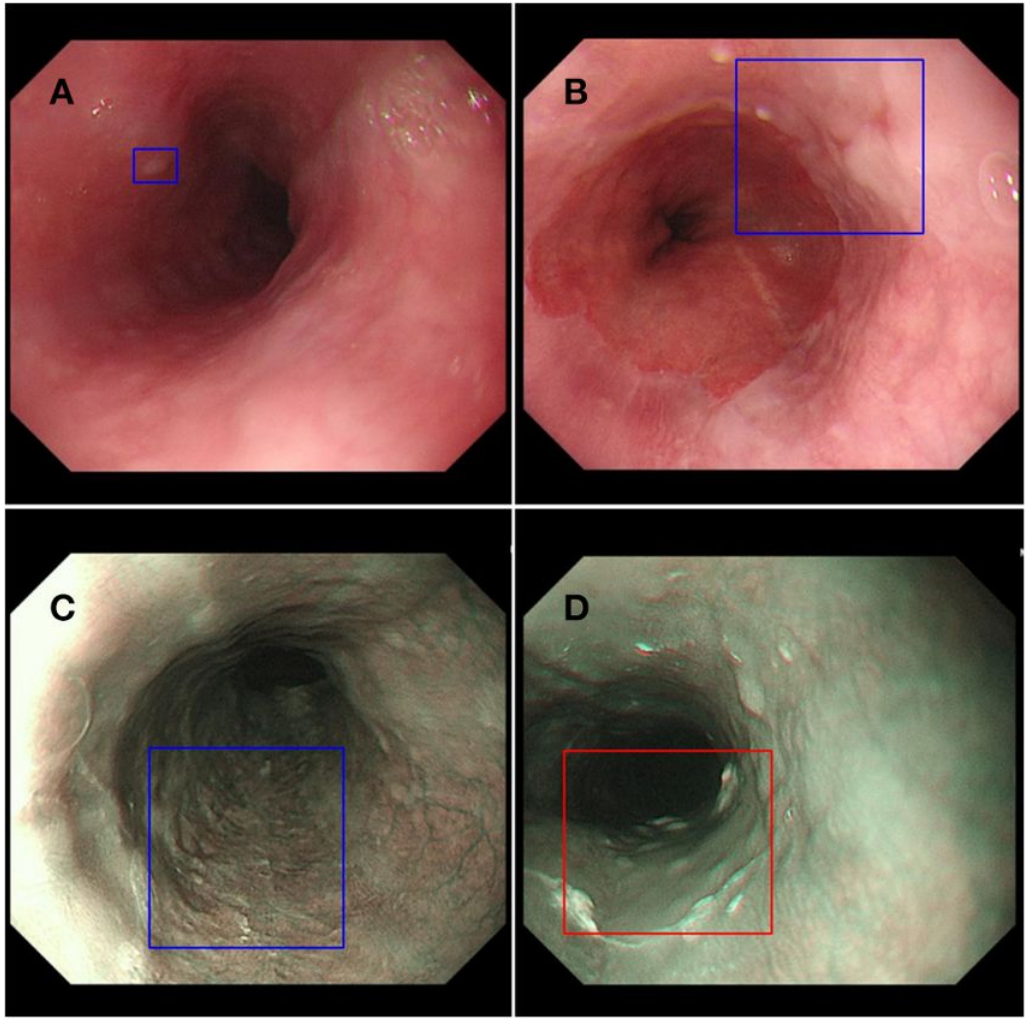

LOW
HIGH
R
pa
Lunit INSIGHT CXR
Abnormality Score : 86%
Cardiomegaly : Low
Mediastinal Widening : Low



Computer Vision for Cancer Detection 2010s

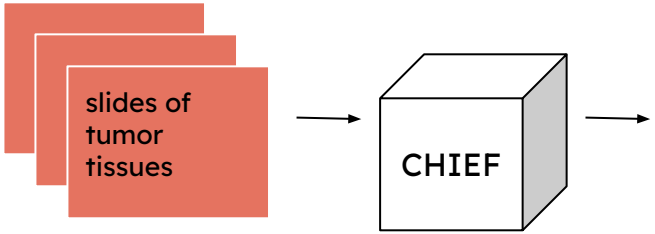
- Gastroenterology has been using AI for colonoscopies. It has been successful in identifying malignant/benign colon polyps and pancreatitis/pancreatic cancer.




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- predicting patient outcomes
- identifying the presence of genes and DNA patterns related to treatment response in tumor micro-environment

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A pathology foundation model for cancer diagnosis and prognosis prediction

Xiyue Wang, Junhan Zhao, Eliana Marostica, Wei Yuan, Jietian Jin, Jiayu Zhang, Ruijiang Li, Hongping Tang, Kanran Wang, Yu Li, Fang Wang, Yulong Peng, Junyou Zhu, Jing Zhang, Christopher R. Jackson, Jun Zhang, Deborah Dillon, Nancy U. Lin, Lynette Sholl, Thomas Denize, David Meredith, Keith L. Ligon, Sabina Signoretti, Shuji Ogino, ... Kun-Hsing Yu + Show authors

Nature **634**, 970–978 (2024) | Cite this article

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Abstract

Histopathology image evaluation is indispensable for cancer diagnoses and subtype classification. Standard artificial intelligence methods for histopathology image analyses have focused on optimizing specialized models for each diagnostic task[1,2]. Although such methods have achieved some success, they often have limited generalizability to images generated by different digitization protocols or samples collected from different populations[3]. Here, to address this challenge, we devised the Clinical Histopathology Imaging Evaluation Foundation (CHIEF) model, a general-purpose weakly supervised machine

CHIEF Model 2024 for

More specific: Single-cell RNA Sequencing

Bulk sequencing cancer DNA & RNA
- average of all tumor cells
- difficult to target tumors

Single cell RNA sequencing
- higher resolution diagnosis
- individual cell level targetted
- more lasting drug responses

